

⚡ Lab 05: Cloud Run Functions Qwik Start - Console

Google Cloud Associate Cloud Engineer | GSP081

Overview

Created, deployed, and tested a Cloud Run function using the Google Cloud Console. Cloud Run functions are serverless and event-driven — they only run when triggered, making them cost-efficient for tasks that don't need to run continuously.

What are Cloud Run Functions?

Feature	Detail
Serverless	No servers to manage — Google handles infrastructure
Event-driven	Only runs when triggered by an HTTP request, Pub/Sub message, or file upload
Trigger used	HTTPS — function triggered by an HTTP request
Cost	Pay only when the function runs — not for idle time

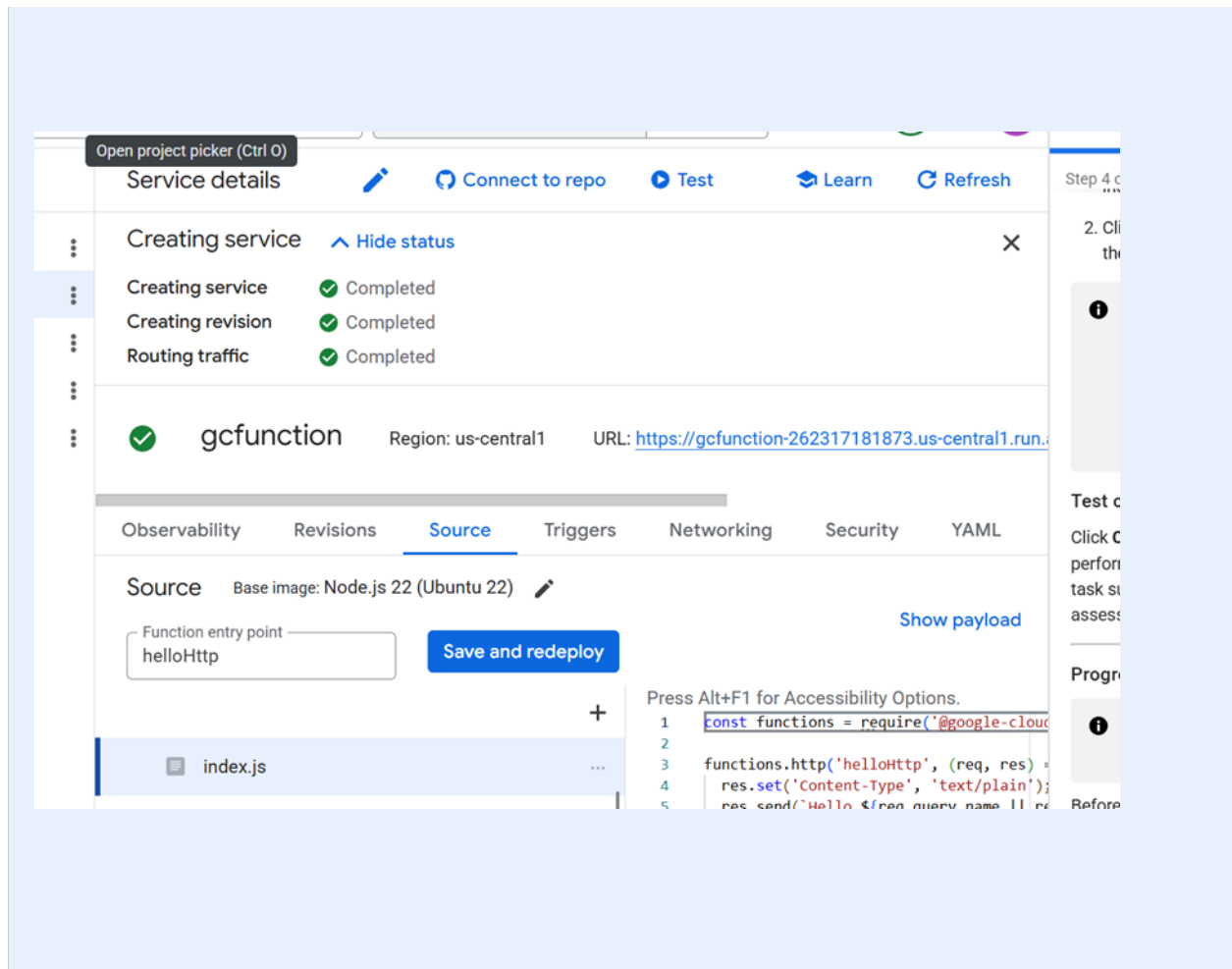
Objectives

- Create a Cloud Run function via the console
- Deploy the function
- Test the function with a triggering event
- View function logs

Task 1 - Create a Function

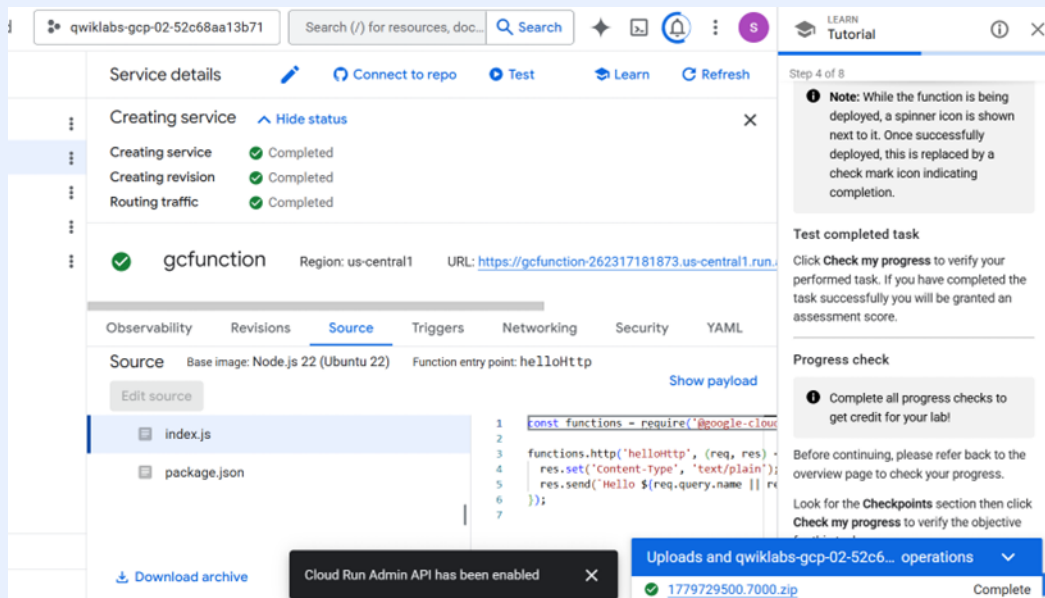
Created a Cloud Run function named gcfuction using the console with the following settings:

- Service name: gcfuction
- Region: us-east1
- Authentication: Allow public access
- Execution environment: Second generation
- Maximum instances: 5



Task 2 - Deploy the Function

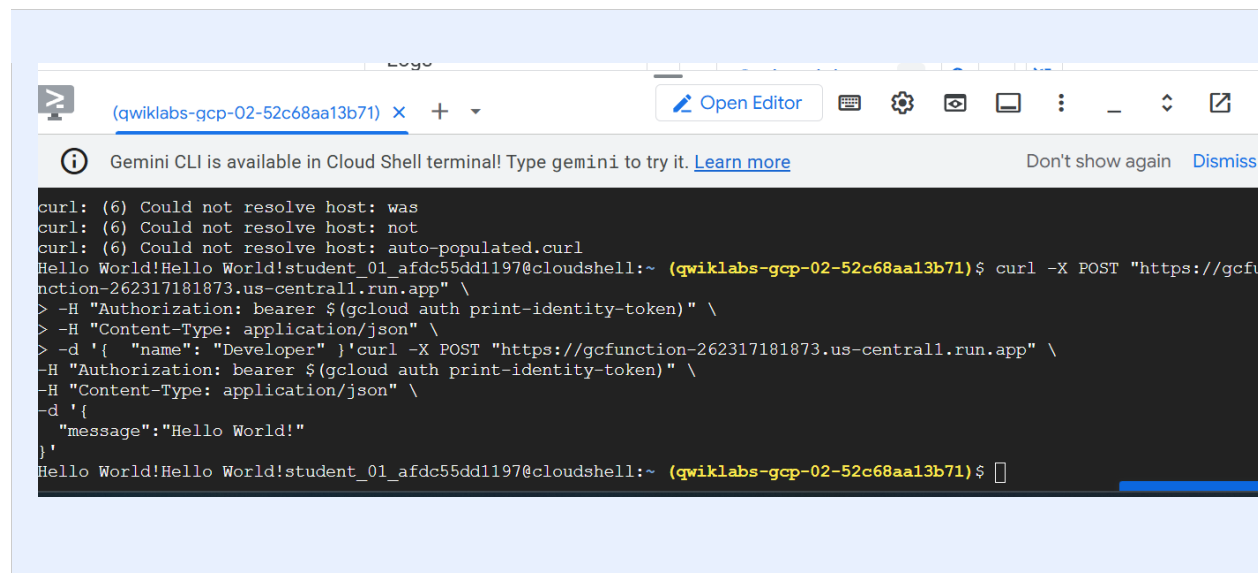
Used the default helloHttp function in the inline editor and clicked Save and Redeploy. Waited for the spinner to become a green checkmark confirming deployment.



Task 3 - Test the Function

Tested the function using the built-in TEST tab with a Hello World triggering event.

- Triggering event: {"message":"Hello World!"}
- Output: Hello World! returned successfully



Task 4 - View Logs

Viewed function logs via Observability → Logs on the Service Details page to confirm the function ran successfully.

The screenshot displays the Google Cloud Observability console interface. At the top, the URL <https://gcf.function-262317181873.us-central1.run.app> is shown, along with the scaling configuration: "Scaling: Auto (Min: 0, Max: ...)". Below this, a navigation bar includes tabs for "Observability", "Revisions", "Source", "Triggers", "Networking", and "Security". The "Observability" tab is active, and within it, the "Logs" section is selected. A sidebar on the left contains links for "Metrics", "Logs", "SLOs", and "Errors". The main content area shows a "Logs" view with a "Severity" dropdown set to "Default". A search bar labeled "Filter" with the placeholder "Search all fields and values" is present. Below the search bar is a table of log entries. The table has columns for "Severity", "Time", and "Summary". The log entries show a sequence of successful function runs, each with a status of "Cloud Run" and a response of "ReplaceService". The bottom of the interface features a toolbar with icons for "Open Editor", a keyboard icon, a settings gear, a view icon, a window icon, and a search icon.

URL: <https://gcf.function-262317181873.us-central1.run.app> Scaling: Auto (Min: 0, Max: ...)

Observability Revisions Source Triggers Networking Security

Metrics

Logs

SLOs

Errors

Logs

Severity: Default

Filter Search all fields and values

Severity	Time	Summary
> i	2026-05-25 13:19:17.994 EDT	Cloud Run ReplaceService
> i	2026-05-25 13:19:20.092 EDT	Cloud Run ReplaceService
> i	2026-05-25 13:29:03.132 EDT	POST 200 147 B
> i	2026-05-25 13:29:14.277 EDT	POST 200 147 B
> i	2026-05-25 13:29:14.324 EDT	POST 200 147 B
> i	2026-05-25 13:30:01.760 EDT	POST 200 147 B
> i	2026-05-25 13:30:01.807 EDT	POST 200 147 B

i No newer entries found matching the current filter.

Open Editor

Real World Application

- Auto-generate thumbnails when images are uploaded to Cloud Storage
- Send push notifications when a new Pub/Sub message arrives
- Process form submissions from a website without managing a server
- Run scheduled data cleanup tasks triggered by Cloud Scheduler

What I Learned

- Cloud Run functions are serverless and event-driven
- Functions only run when triggered — no cost when idle
- HTTPS is one trigger type — others include Pub/Sub and Cloud Storage
- The inline editor lets you write and deploy code directly in the console
- Logs are available in the Observability tab of the service details page
- Second generation execution environment offers better performance

Challenges & How I Solved Them

- On the Create function page accidentally selected Deploy a container instead of Write a function. This took me to a different setup flow that did not match the lab instructions. Had to restart the entire lab to get a fresh environment and start over from the beginning.
- Key lesson: Always read the options carefully before clicking on the Create function page. Write a function and Deploy a container are two different paths and choosing the wrong one requires restarting the lab since you cannot switch mid-way.

Resources

GCP Cloud Run Functions Docs: <https://cloud.google.com/functions/docs>
Lab ID: GSP081